

TESO INTEGRATED SECONDARY SCHOOL- NGORA

S.5 END OF TERM 2 ASSESSMENT 2025

PRINCIPAL MATHEMATICS PAPER 2

TIME: 2 HOURS AND 30 MINUTES

INSTRUCTION: Attempt 4 items in all

ITEM ONE

In research conducted by the national agricultural research institute, scientists are studying whether there is relationship between soil fertility scores and crop yield on various farm plots. Soil fertility was measured on a scale of 0 to 100, and crop yield is measured in kilograms per hectare. A total of 12 sample plots were tested. The team believes that higher soil fertility should result in higher yields, but requires statistical proof. A conclusion will be made if the correlation of the results is moderate or above. Unfortunately, 2 of the plots had incomplete data due to equipment malfunctions. Plot 4 is missing crop yield reading and plot 9 is missing a soil fertility score.

The data is as follows:

Plot No.	1	2	3	4	5	6	7	8	9	10	11	12
Soil Fertility Score	75	60	85	90	55	70	68	88		62	80	66
Crop Yield (kg/hectare)	78	58	92		54	74	72	90	70	60	85	69

Task:

Help the scientist;

- find the missing value for the sample plots 4 and 9.
- Can the institute reasonably conclude that higher soil fertility leads to higher yield?

ITEM TWO

A group of learners were taken to a certain factory in Jinja to fully understand the concept of probabilities. The factory produces light bulbs using three machines: Machine A, Machine B, and Machine C.

- Machine A produces 40% of the total output, and 3% of its bulbs are defective.
- Machine B produces 35% of the total output, and 3% of its bulbs are defective.
- Machine C produces 25% of the total output, and 2% of its bulbs are defective.

A bulb is selected at random from the factory's output.

Later, these learners were asked to use the knowledge to also analyse the given problem.

Two boxes M and N contain 4 red and 3 blue pens and 3 blue and 4 red pens respectively. A box is selected at random, and from it a pen is drawn and placed into the second box, the

second box is shaken, and from it a pen is drawn and placed into the first box.

Task:

- Determine the probability that the selected bulb was produced by Machine A and is defective.
- Given that the selected bulb is found to be defective, calculate the probability that it was produced by Machine B.
- Determine the probability that the pen selected from the first box is red.

ITEM THREE

Two vehicles, a lorry and a car, have been observed at Masaka road. Both are moving along a straight, horizontal road. A lorry starts from Nyando moving with a constant acceleration of 2 ms^{-2} . At the same instant, a car moving with a speed of 20 ms^{-1} and a constant acceleration of 30 ms^{-2} is 400 m behind Nyando and moving in the same direction as the lorry.

The lorry reaches Lwara for oil refilling, the distance between Lwara and Buwunga is $6x \text{ m}$, the lorry starts from Lwara and accelerates uniformly to a speed of $v \text{ ms}^{-1}$, covering distance $x \text{ m}$. The lorry maintains this speed until it has travelled a further $3x \text{ km}$, and it then retards uniformly to rest at Buwunga. The total time taken to travel from Lwara to Buwunga is T .

Task:

- Determine how far from Nyando the car overtakes the lorry.
- Determine the speed of the lorry when it is being overtaken.
- If T is the total time from Lwara to Buwunga, show that $T = \frac{9x}{v}$ seconds.

ITEM FOUR

As part of a performance review, a logistics company organized a practical assessment for a group of 180 trainee dispatchers. The aim was to evaluate their speed and accuracy in handling simulated delivery assignments. Their scores out of 100 were compiled and grouped as shown below.

Marks	<20	<25	<35	<45	<55	<60	<65	<80	<95
Cumulative number of learners	0	6	24	66	110	130	154	174	180

Two additional policies were announced during the assessment:

- Any dispatcher scoring 75 or above would qualify for a performance bonus.
- Those who scored below the average would be enrolled in a refresher training program.

Further guidelines include;

- Only 20 dispatchers were to receive a “top performance recognition”.
- If the measure of spread in scores (standard deviation) exceeded 10 marks from the

mean, the assessment would be flagged as inconsistent.

Task:

- a). Assist the operations manager in determining:
 - i) How many dispatchers will receive the performance bonus.
 - ii) How many must be enrolled in a refresher training program.
 - iii) The minimum score required to be among the top 20 performers.
- b) Based on the scores, would you consider the performance results consistent?

ITEM FIVE

A new electronic component has a lifespan denoted by the random variable X can vary. Due to its unique design, the probability density function for its lifespan is given as;

$$f(x) = \begin{cases} k(x + 2), & -1 \leq x \leq 0 \\ 2k(1 - x); & 0 < x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

where k is a constant. Amulen bought one item from the company producing these components.

Task:

- a) Sketch the probability graph of the function showing the distribution of the components lifespan.
- b) Determine the value of the parameter k and evaluate the mean lifespan of this component.
- d) Given that the component has lasted longer than 0 units of time, what is the probability that its lifespan will be less than 0.5 units of time $\left(\frac{0 < x < \frac{1}{2}}{x > 0}\right)$.

END